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Comparative study of regressor and classifier with decision tree using modern tools

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ABSTRACT

Machine Learning is one of the importantareas for modeling the data and itcan be saidthat this is the core part of the field of Data Science. Supervised Machine Learning (SML)has many algorithms to train the machine. The decision tree as the classifier is used to trainthe model based onthe categorical label and the Decision Tree as Regressor is used to trainthe model based ona non-categorical label. There are two kinds of algorithms like Classification and Regression. In this paper, we focus on the Decision Tree as a Regressor and Classifier and compare the metrics. This paper describes the decision tree with the analysis as well as a comparison with the most efficient algorithm based on the different datasets using python programming. Results show the accuracy and comparison of the decision tree as a Regressor and Classifier. Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Confusion Matrix are performance parametersused to analysis of decision treesand also using different python libraries to analyze and visualize the result.In this paper, we used a shopping mall dataset for classification as a case study from UCI (Machine Learning Repository) to predict that users purchase an item or not. This dataset contains 400records. Decision Tree as a Regressor is used as the dataset from the Kaggle repository for analysis and visualization of results and show comparison. In this paper, the Accuracy score is the most important measure to compare decision treesbased on regression and classification but Mean Squared Error (MSE) is also an important factor to decide and split the node into two or more nodes.
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1. Introduction

1.1. Decision tree

A decision tree [1] is an algorithm for learning which is important and mostly used to analyzing of data and a decision tree can perform regression and classification problems. The motive to study the decision tree is simple. A decision tree contains decision nodes (Test the value of an attribute), edges (Outcome of a test and connect with next node), and leaf node (Predict the outcome) combined and make a complete structure of decision tree as shown in Fig. 1.

In the decision tree, each attribute of a dataset is treated as a node where a special and unique node is a root node. To elaborate the working of the decision tree, we must start with a unique node and elaborate down the tree to satisfy our parameters and decision. This technique will be further progress until a terminal node is encountered. The terminal node would bethe prediction or the output of the decision tree. Consider a situation in Fig. 1 where an employee asks for your laptop. Some situationshelp to take an appropriate decision.

a. Is an employee your close colleague or just a colleague? If an employee is just a colleague then refuse the request and then go to the next step.
b. Is an employee asking for the laptop the first time? If yes then give the laptop to your employee otherwise go to the next step.

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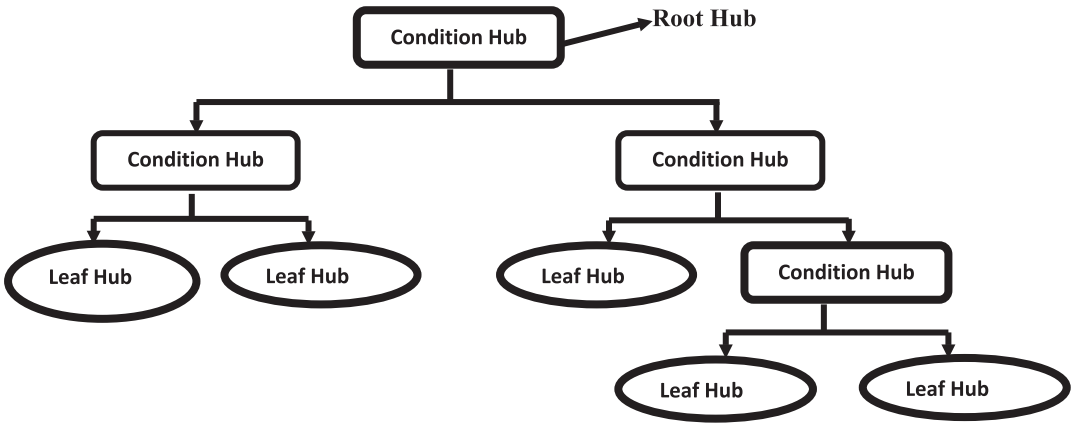

Fig. 1. Decision Tree Structure [2].

c. Was the scratch or damaged the laptop last time? If yes then refuse the request otherwise give the laptop to your colleague.

We must use a decision Tree because it generally follows the human process that can understand data and interpretation of data efficiently. It is a way by which you can see the rationale for the information to decipher.

2. Elements of a decision tree

Every decision tree consists of these elements [2]:

a. Node: It is a position from where the tree will be splits based on the independent variable and its value of the dataset.
b. Edge: It is shown the direction of the decision of a split to the next node.
c. Root: This is the first node where the first split takes place.
d. Leaf Node: It is also called Terminal Node that predicts the result of the decision tree.

3. Selection of attribute selection measure (ASM):

To create a Decision Tree, The important goal is to select an appropriate attribute from the list of features of a dataset for the root node as well as the decision node. This is achieved with the help of a technique which is known as ASM (Attribute Selection Measures) [2]. There are two methods to select appropriate attributes from a feature list of the dataset (see Figs. 2–6).

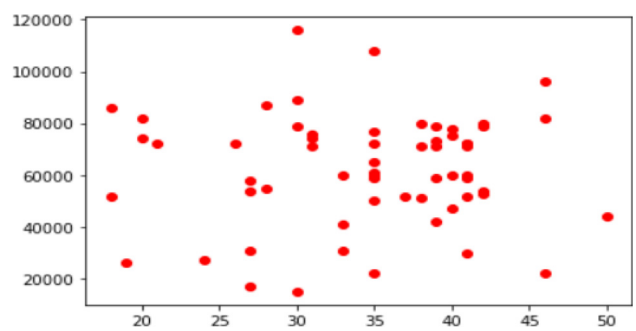

Fig. 3. Customers who are not purchasing.

a) Information Gain: After splitting the dataset based on an attribute, we can changes the entropy value of the attribute is known as Information Gain.Utilizing the information Gain, we can characterize the hub and make the decision tree. Hub, the most noteworthy information of data acquired, is being parted first and the decision tree attempts to build the worth of data acquired. By this formula [2], we can calculate information gain:

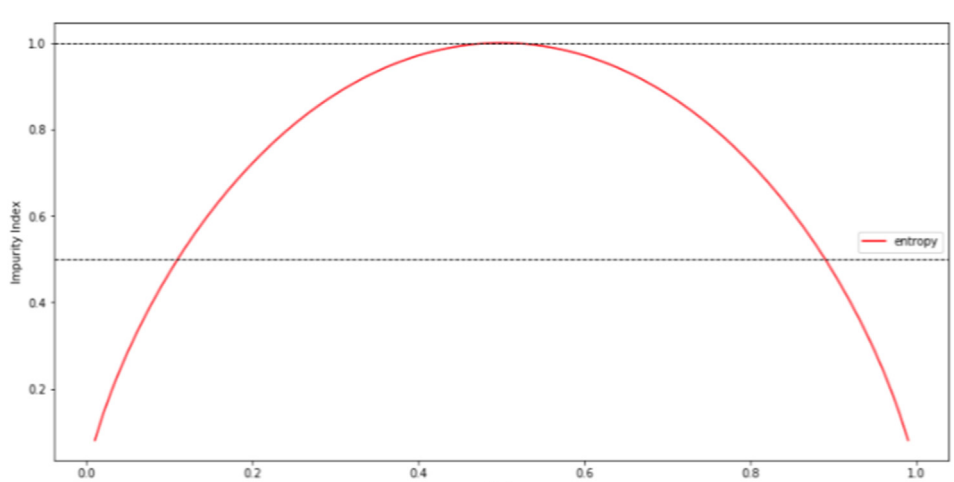

Fig. 2. Probability Vs Impurity Index of Entropy [4].

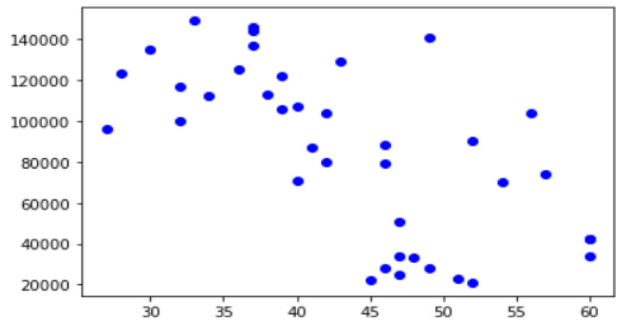

Fig. 4. Customers who are purchasing.

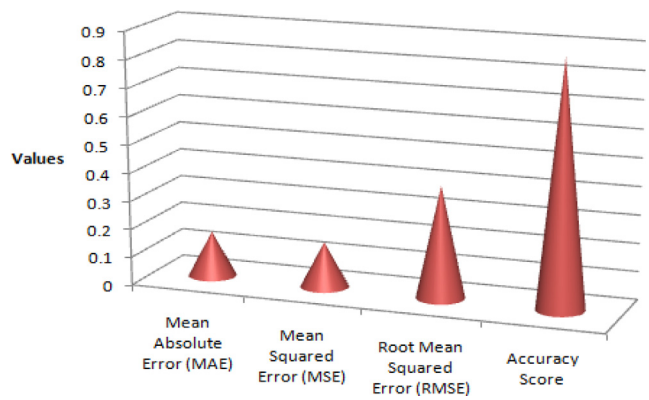

Fig. 5. Performance Parameters of Decision Tree as a Classifier.

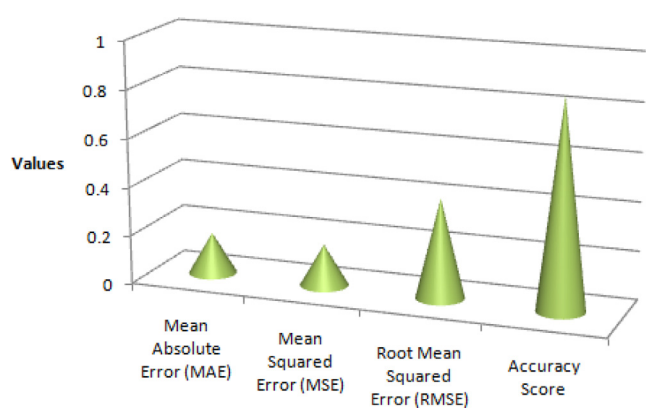

Fig. 6. Performance Parameters of Decision Tree as a Regressor.

$$\text{Information gain} = \text{S} - [(\text{Weighted Average}) * \text{Entropy (Each Input)}] \quad (1)$$

where S is the total number of samples of a dataset

Some Information Gain based calculations are created by Ross Quinlan which are ID3 and C4.5 [3]. These calculations foster decision trees based on data acquired from the preparation information.We know that the entropy signifies the uncertainty in the dataset. More uncertainty in the dataset is more entropy. It is used to measure impurity. The entropy is 0 if all attributes at a node belong to the same class, and the entropy is maximal if our class distribution is uniform. We can understand entropy using the visual concept, plot the graph between the impurity index and the probability range which is [0, 1] for first class.

Above all else, we should discover the entropy of each attribute and determined, then, at that point after computing the data acquire for every one of the qualities by utilizing the equation.

$$E(S) = -P(yes) * \log_2 P(yes) - P(no) * \log_2 P(no) \quad (2)$$

where,

P(yes) = probability of yes

P(no) = probability of No

The attribute which hasthe highest information gain is denoted as the most informative attribute and we can select that it is the root node. And this process is repeated until all the attributes of the dataset are in the decision tree.

b) Gini Index [4]: It is a metric used to measure of impurity/purity of an attribute in a dataset. This would be used while creating a decision tree using the CART supervised learning algorithm. The Gini index is used to calculate an attribute. Whenever it is low or high then we will prefer of low Gini index attribute. Generally, the Gini Index is used for binary spilt so that the CART algorithm is used to create a decision tree through the binary split. It can be calculated by using the formula:

$$Gini\ \ Index = 1 - \sum_{j=1}^{n} P_j^2 \quad (3)$$

where, $P_j$is the probability

3.1. Regression

Regression [5] can be partitioned into straightforward direct regression and various straight regression. Regression is examining the connection between autonomous variables and ward variables. We can assemble the regression model of ward variable y and the autonomous variable *x* that is ascertaining the variable y and foresee the pattern of y. The simple linear regression can be calculated as:

$$y = b_0 + b_1 * x + e$$

where,

b_0 is the intercept on the y-axis or constant

b_1 is the slope of line or coefficient of the predictor variable x

e is the residual error

When we have multiple variables x_1 and x_2, the regression equation can be shown as:

$$y = b_0 + b_1 * x_1 + b_2 * x_2 + e$$

If we have a large number of variables, means multivariate regression, in this case, we use well- known regression model that is ridge regression and lasso regression.

In the regression, the independent variable is also called the input variable or features, and the dependent variable is also called the target variable or output variable or label.

There are some methods to measure the regression models. These measurements are commonly used to evaluate the performance of a regression.

- Mean Squared Error (MSE): The mean or average of the squared differences between predicted and actual values is called MSE.

$$MSE = \left(\frac{1}{N}\right) * (y_{actual} - y_{predict})^2$$

- Root Mean Squared Error (RMSE): The square root of the error is calculated which means that is RMSE.

$$RMSE = \sqrt{MSE}$$

- Mean Absolute Error (MAE): It is calculated as the mean of the absolute error values. So, the difference between an actual and predicted value may be positive or negative and is forced to be positive when calculating the MAE.

$$MSE = \left(\frac{1}{N}\right) * abs(y_{actual} - y_{predict})$$

3.2. Classification

It is a process to predict the class [6] and select a classifier based on the independent variable. For example, to select T-shirt sizes like XL, XXL, M of a customer based on the independent variable like customer height, customer weight. The frequently used classification algorithms are Logistic Regression, Decision Tree, Naïve Bayes classifier, KNN, Support Vector Machine (SVM), and Neural Network. Logistic regression helps us to predict the salary of an employee and using a decision tree algorithm to predict the outcome that play is possible or not today. It is a Supervised Learning measure where yield is having objective worth (discrete worth). It tends to be either a twofold or multi-class arrangement. In double grouping, it tends to predict either 0 or 1; yes or no yet in the event of multi-class arrangement, it tends to predict more than one class. For instance, Gmail groups sends in more than one class like social, advancements, spam. In this, a downright yield y is planned to enter variable(x).

$$y = f(x)$$

where, y is categorical output

When we complete our model using classification, evaluate its performance using different ways. In these ways, the most significant technique to assess the execution of the arrangement model is the Confusion Matrix. It gives a framework/table as yield to discover the presence of a model. It is additionally called Error Matrix.The confusion matrix provides correct and incorrect predictions in a summarized form.

By Table 1, we can find the correct predictions to the sum of diagonal elements i.e. Yes Positive (YP) and Yes Negative (YN), and find incorrect predictions to the sum of off-diagonal elements i.e. No Negative (NN) and No Positive (NP). Mathematically, we can find the accuracy of the classification model using confusion matrix by this formula:

$$Accuracy = \frac{(YP + YN)}{Total\ \ Polulation}$$

where, Total Population is the sum of YN, YP, NN, and NP

In this paper, we are finding measures using decision tree as regressor and classifier and find knowledge about the improvement of MSE, RMSE, MAE, an accuracy score. We know that different algorithms are using to classify the data and provide measures but our methodology is a decision tree to analyze the measures and findsthe best solution (see Tables 2–5).

3.3. Decision tree as classifier

The decision tree is also known as Classification and Regression Tree [7] because it is used for both types of problems. Mostly decision tree algorithm is used for classification problems but when required then can use for regression problems also. A decision tree is used to predict the dependent variable from the independent variable. Decision Tree used as a classifier to predict the value. Decision Tree is an advanced and innovative technology that is used to solve science & engineering problems. It is a non-parametric approach. It deals with high dimensionality taking variables on its high priority. A Decision tree as a classifier is included node, branch, root, leaves. We have incorporated a shopping mall dataset as a case study from UCI (Machine Learning Repository) to predict that users purchase or not. This dataset contains 400 records with the features User ID, Gender,Age, EstimatedSalaryand Purchased.There are some steps to classify the dataset using a decision tree:

Table 1
Structure of confusion matrix.

	Real Positive	Real Negative
Anticipated Positive	Yes Positive	No Positive
Anticipated Negative	No Negative	Yes Negative

3.4. Import the libraries

In this paper, we will use Python3.6 to set up a decision treeas a classifier forthe analysis model. We introduce the open-source language Python 3.6 and import all the libraries for machine learning algorithms. Moreover, we will also importthe Pandas library and convert the dataset into a more advanced data structure i.e. DataFrame.

3.5. Data source

In this paper, we proposed a dataset of 400 records and 5 features. To convert a dataset of .CSV file into DataFrame using Pandas library. Here, DataFrame is the 2D data structure of Pandas library.

Presently, to anticipate if a client will buy the item, one necessity to discover the connection between Age and Estimated Salary. Here User ID and Gender are not significant components for discovering this. Using python, we can select features and labels.

```
#Features
x = dataframe[['Age','EstimatedSalary']] or dataframe.iloc
[:,2:4].values
#Labels
y = dataframet['Purchased']
```

After that,we will part the dataset into train information and test information. Here we split 75% of the information for preparing the model and 25% of information to test the presentation of our model.

```
x_train, x_test, y_train, y_test = train_test_split(x, y,
test_size = 0.25)
```

In a DataFrame, the Total number of 400 records is divided into train data (75%) of 300 records and test data (25%) of 100 records.

Perform highlight scaling here because Age and Estimated Salary highlights esteem lie in various reaches implies that huge hole between them.If we donot do the scalingof the Estimated Salary and Age features then the Estimated Salary feature will be strong when compared to the Age feature.

```
scale_x = MinMaxScaler()
x_train_s = scale_x.fit_transform(x_train)
x_test_s = scale_x.transform(x_test)
```

Here, Age and Estimated compensation highlights esteems are scaled and presently there in the scope of 0 to 1. Subsequently, each element will contribute similarly in dynamic.

Hence, we are training our Decision Tree model as a classifier using the library in python and then fit the model with training data of feature and label. After that,we can predict test values using test data of features.

Table 2
DataFrame.

S.No	User ID	Gender	Age	EstimatedSalary	Purchased
0	13,625,313	M	20	18,020	0
1	13,820,034	M	33	21,030	0
2	13,668,655	F	27	42,040	0
3	13,653,176	F	26	57,030	0
4	13,834,282	M	20	76,040	0
5	13,788,343	M	28	58,060	0
6	13,568,834	F	28	84,070	0
7	13,644,619	F	30	150,080	1
8	13,670,495	M	26	33,070	0
9	13,737,201	F	36	65,080	0

Table 3
DataFrame of actual data and predict data using decision tree as classifier.

S.No	y_test	y_predict
1	1	1
2	1	0
3	0	0
4	1	1
5	0	0
–	–	–
95	1	1
96	1	1
97	0	0
98	1	0
99	0	0

Table 4
DataFrame of actual data and predict data using decision tree as regressor.

S.No	y_test	y_predict
1	0.0	0.0
2	0.0	0.0
3	1.0	1.0
4	0.0	0.0
5	0.0	1.0
6	0.0	0.0
7	1.0	1.0
8	0.0	0.0
9	1.0	1.0
10	0.0	0.0
11	0.0	0.0

Table 5
Evaluation metrics for decision tree.

Algorithm	Parameters	Evaluation Parameters	Values
Decision Tree as Classifier	Independent Parameter: Age, EstimatedSalary	Mean Absolute Error (MAE)	0.16
		Mean Squared Error (MSE)	0.16
		Root Mean Squared Error (RMSE)	0.40
		Accuracy Score	0.87
Decision Tree as Regressor	Target Parameter: Purchased	Mean Absolute Error (MAE)	0.17
		Mean Squared Error (MSE)	0.17
		Root Mean Squared Error (RMSE)	0.41
		Accuracy Score	0.86

```
model_dt = DecisionTreeClassifier()
model_dt.fit(x_train_scaled, y_train)
y_predict = model_dt.predict(x_test_scaled)
```

So, we can find predicted values with respect to actual values of test data from the dataframe and create a new dataframe for actual and predicted values.

Before that, we can visualize the data using a scatter plot of the MatPlotLib library in python from our dataset and find that how many customers are purchasing or not.

Decision tree as a classifier is classified the data according to its features and then we can calculate the accuracy of our model where we used two features (Age, Estimated Salary) which are 0.87 (87%). When we used three parameters (Gender, Age, Estimated Salary) as features then our model accuracy will be 0.51 (51%) which is not good because the Gender attribute is not useful to predict that customer is purchased or not.

We can test the performance [8] of our model using Confusion Matrix. A Confusion Matrix is a table that is utilized to assess the performance of an older model. You can likewise envision the presentation of an Algorithm. The essential of a Confusion Matrix is the quantity of right and wrong expectations is summarized class-wise. Inclining esteems address precise expectations, while non-slanting components are mistaken forecasts. In the yield, 53 and 31 are genuine forecasts, and 10 and 6 are inaccurate expectations. The Confusion Matrix is:

$$\begin{matrix} 53 & 10 \\ 6 & 31 \end{matrix}$$

3.6. Decision tree as regressor

The decision tree as a regressor is included node, branch, root, leaves. We have incorporated a shopping mall dataset as a case study from UCI (Machine Learning Repository) to predict that users purchase or not. This dataset contains 400 records with the features User ID, Gender, Age, EstimatedSalaryand Purchased.

As a classifier, the Decision Tree also works as a regressor. For regressor, we are using another python library that is DecisionTreeRegressor of sklearn package. Here, we are training our Decision Tree model as a regressor using the library in python and then fit the model with training data of feature and label. After that, we can predict test values using test data of features.

```
model_dt = DecisionTreeRegressor()
model_dt.fit(x_train_scaled, y_train)
y_predict = model_dt.predict(x_test_scaled)
```

So, we can find predicted values with respect to actual values of test data from the dataframe and create a new dataframe for actual and predicted values using the decision tree as a regressor.

Decision tree as a regressor is predicted the data according to its features and then we can calculate the accuracy of our model where we used two features (Age, Estimated Salary) which are 0.90 (90%). When we used three parameters (Gender, Age,

Estimated Salary) as features then our model accuracy will be 0.55 (55%) which is not good because the Gender attribute is not useful to predict that customer is purchased or not. We can test the performance of our model using Confusion Matrix. The Confusion Matrix is:

61	5
5	29

In the output, 61 and 29 are actual predictions, and 5 and 5 are incorrect predictions.

3.7. Comparision and result analysis

A decision tree as a supervised machine learning algorithm is used as regressor or a classifierto develop the models in the form of a tree structure. A decision tree is a work with breaks down a dataset into different branches and leaves. Each node of a decision tree has two or more branches, each branch evaluates the value of the attribute. Leaf node represents the target value that we want to predict. Mean Squared Error (MSE) [10] is the important factor to decide and split the node into two or more nodes. Evaluation metrics of Regressor and Classifier of Decision Tree are shown in the table.

4. Conclusion

Now, after evaluating the Decision Tree on the dataset, we compare the both methods regarding performances based on two important metrics Accuracy Score and Mean Squared Error(MSE) to find that the decision tree as a classifier is better than the regressor.In this paper, we have analyzed and compare the performance parameters and use of decision tree as Classisfier and Regressor [9]. We analyze the results using various parameters like MAE, MSE, RMSE, an accuracy score. In our proposed work, we found appropriate results. We are also analyzing the effect of the number of features on the selected algorithm. At last, we found the outcome of this paper is that the Decision tree as a classifier is better than the regressor based onkey performance parameters.

We can use algorithms on a large volume of datasets and a variety of independent variables. Our system compares the performance in terms of parameters like MAE, MSE, RMSE, Accuracy score. More parameters can be used to evaluate the performance of the proposed algorithm in the future.

Finally, our proposed work is useful to the students, researchers, academicians, and others and helpful to enhance their previous work.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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